Evaluate 2:

Calculate the pH of a 2 L solution containing 10 mL of 5 M acetic acid and 10 mL of 1 M sodium acetate. The pK_a for acetic acid is 4.76.

Evaluate 3:

The pK_a of the phosphate buffer system ($[H_2PO_4^-]/[HPO_4^{-2}]$) is 6.8. What are the relative concentrations of $[H_2PO_4^-]$ and $[HPO_4^{2-}]$ in a urine sample that has a pH of 4.8.

1.5.4 Uses of Buffers

- Buffer solutions are necessary to maintain the pH range of biological fluids. A buffer of carbonic acid ($H_2CO_3^-$) and bicarbonate (HCO_3^-) is present in blood plasma, to maintain a pH between 7.35 and 7.45.
- Enzymes are functional only under narrow pH range.
- In Industries, buffer solutions are used in fermentation processes and in setting the correct conditions for dyes used in coloring fabrics.
- They are also used for the calibration of pH meters.
- Buffers are mainly used in clinical and research laboratories. Example., phosphate buffered saline (PBS) at pH 7.4.

Determination of pH : pH meter

A pH meter is used to measure the pH of a solution. It consists of a voltmeter attached to a pH-responsive electrode and a reference electrode. The difference in electrical potential produced by hydrogen ions between pH electrode and a reference electrode is measured. A pH meter is utilized in different laboratories and industries to measure pH.

1.5.5 pH and Buffer system in Body fluids

Body fluid

All parts of the body require nutrients and the metabolic wastes produced in them need to be removed from the body. Hence, there is a need to transport various substances like digested food materials, hormones, catabolites, enzymes, various gases from one part of the body to another. These movements are achieved through body fluids. In addition to this, body fluids also provide the medium for the occurrence of metabolic reactions. Water is major component of body fluids. Water is present within and around the cells of the body, and within all the blood vessels. The total body water (TBW) is approximately 60% of body weight.

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